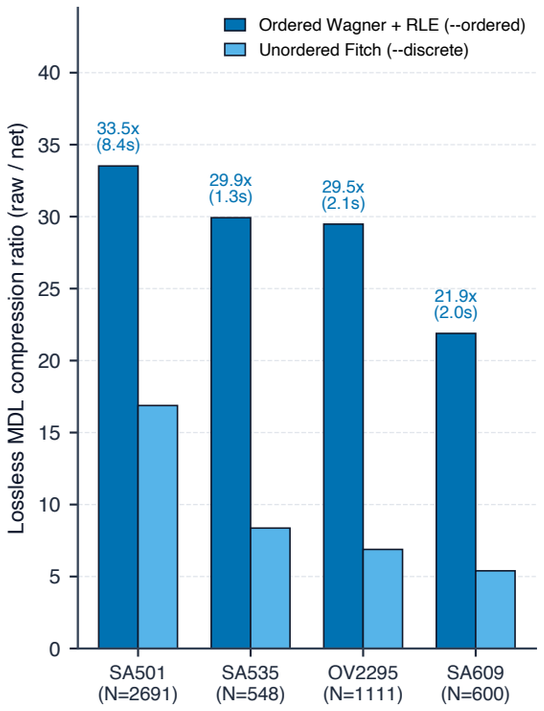
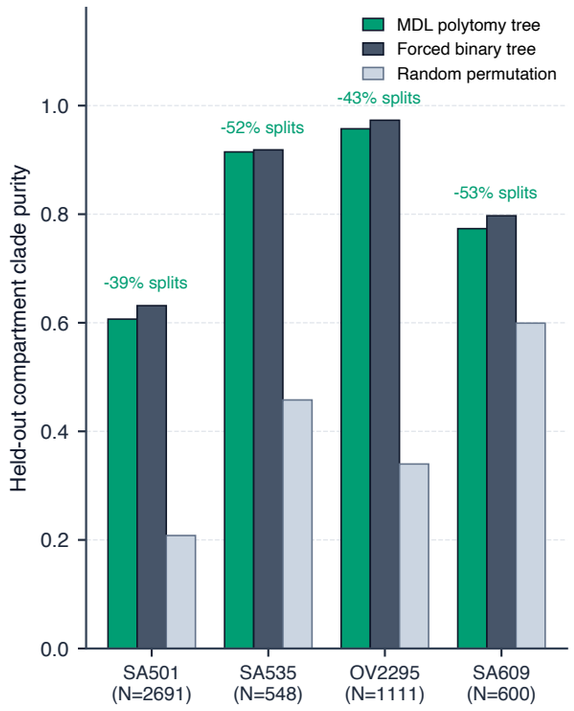
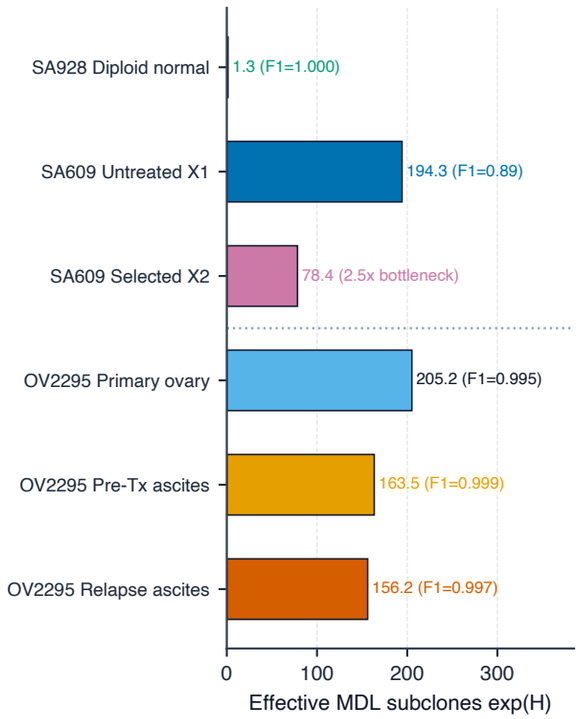
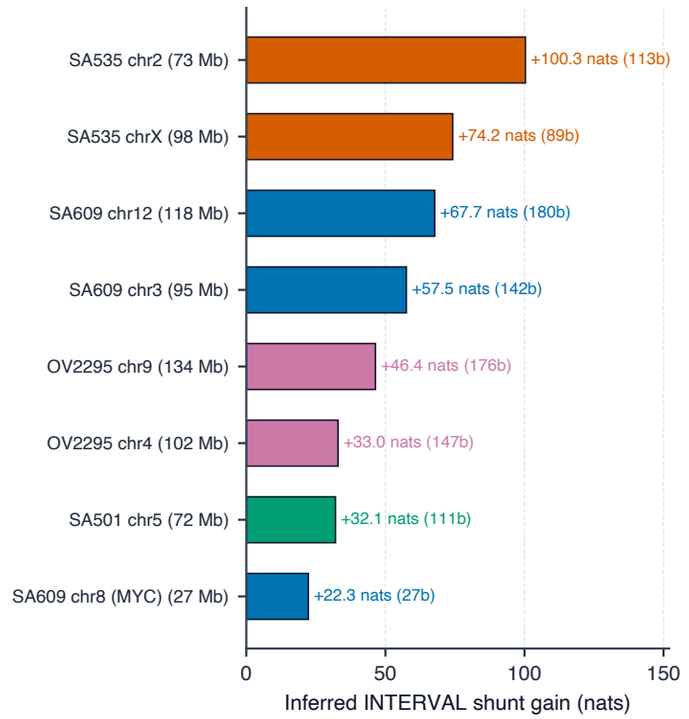
A


scWGS 500-kb CNA compression
(4,950 cells x 4,375 bins, K=12)
Ordered Wagner + RLE (--ordered)
Unordered Fitch (--discrete)
Lossless MDL compression ratio (raw / net)
33.5x (8.4s)
29.9x (1.3s)
29.5x (2.1s)
21.9x (2.0s)
SA501 (N=2691)
SA535 (N=548)
OV2295 (N=1111)
SA609 (N=600)

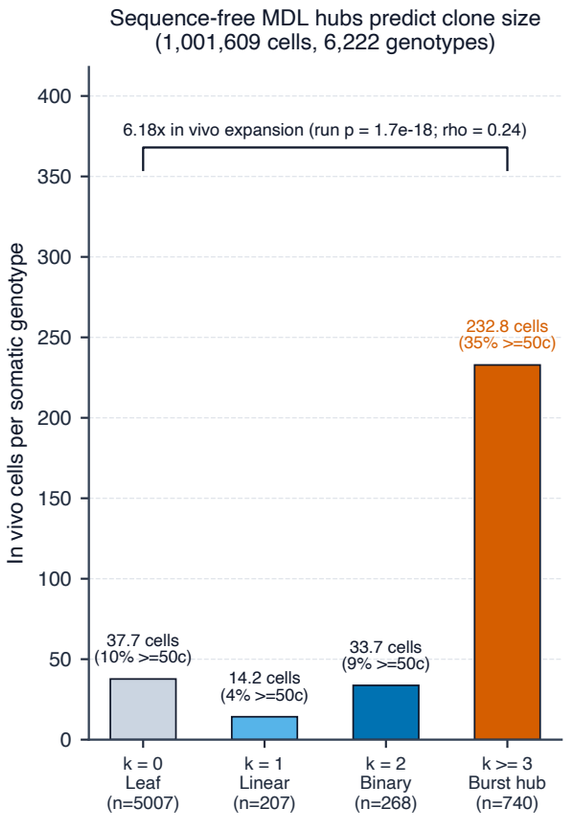

Zero-leakage clade concordance
& false-split contraction (%)
MDL polytomy tree
Forced binary tree
Random permutation
Held-out compartment clade purity
-39% splits
-52% splits
-43% splits
-53% splits
SA501 (N=2691)
SA535 (N=548)
OV2295 (N=1111)
SA609 (N=600)

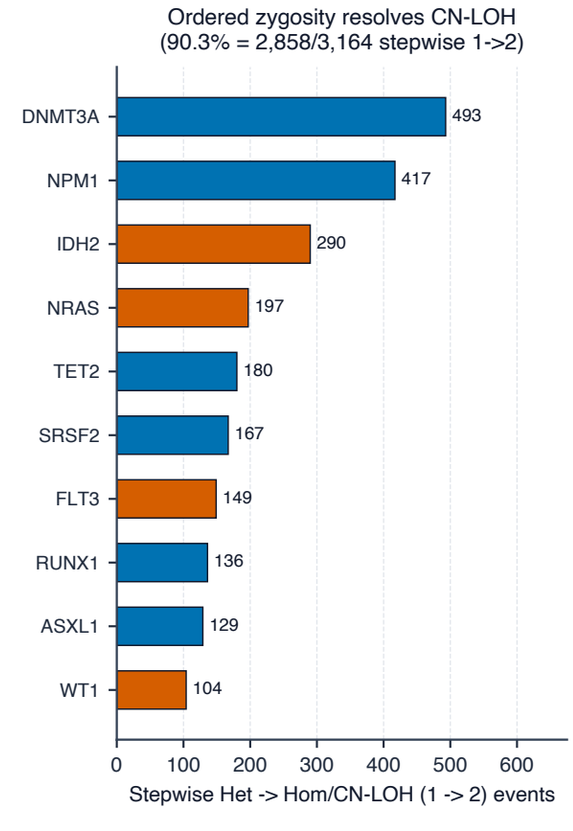
B


Metastatic & selective
subclone diversity
SA928 Diploid normal
1.3 (F1=1.000)
SA609 Untreated X1
194.3 (F1=0.89)
SA609 Selected X2
78.4 (2.5x bottleneck)
OV2295 Primary ovary
205.2 (F1=0.995)
OV2295 Pre-Tx ascites
163.5 (F1=0.999)
OV2295 Relapse ascites
156.2 (F1=0.997)
Effective MDL subclones exp(H)

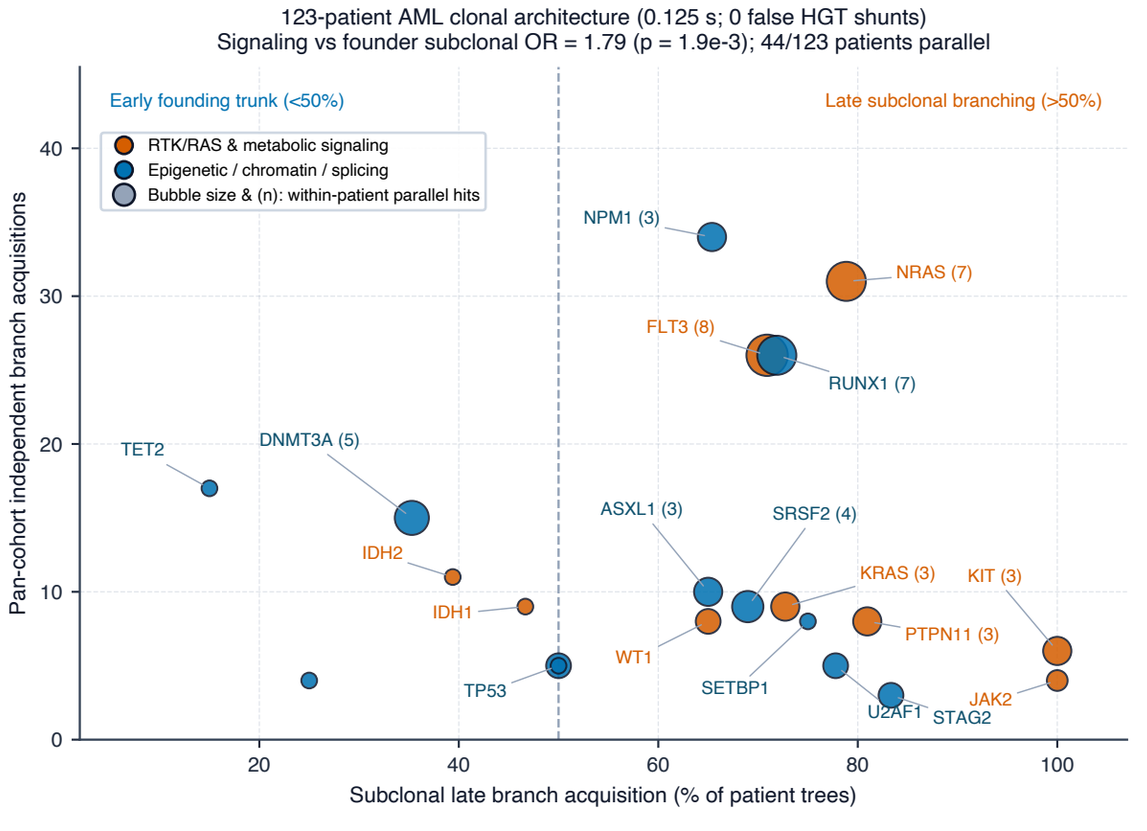

Stage 4 chromosomal-arm
CNA / LOH shunts (P = 1.000)
SA535 chr2 (73 Mb)
+100.3 nats (113b)
SA535 chrX (98 Mb)
+74.2 nats (89b)
SA609 chr12 (118 Mb)
+67.7 nats (180b)
SA609 chr3 (95 Mb)
+57.5 nats (142b)
OV2295 chr9 (134 Mb)
+46.4 nats (176b)
OV2295 chr4 (102 Mb)
+33.0 nats (147b)
SA501 chr5 (72 Mb)
+32.1 nats (111b)
SA609 chr8 (MYC) (27 Mb)
+22.3 nats (27b)
Inferred INTERVAL shunt gain (nats)

C


Sequence-free MDL hubs predict clone size
(1,001,609 cells, 6,222 genotypes)
6.18x in vivo expansion (run p = 1.7e-18; rho = 0.24)
In vivo cells per somatic genotype
37.7 cells (10% >=50c)
14.2 cells (4% >=50c)
33.7 cells (9% >=50c)
232.8 cells (35% >=50c)
k = 0 Leaf (n=5007)
k = 1 Linear (n=207)
k = 2 Binary (n=268)
k >= 3 Burst hub (n=740)


Ordered zygosity resolves CN-LOH
(90.3% = 2,858/3,164 stepwise 1->2)
DNMT3A 493
NPM1 417
IDH2 290
NRAS 197
TET2 180
SRSF2 167
FLT3 149
RUNX1 136
ASXL1 129
WT1 104
Stepwise Het -> Hom/CN-LOH (1 -> 2) events

D


123-patient AML clonal architecture (0.125 s; 0 false HGT shunts)
Signaling vs founder subclonal OR = 1.79 (p = 1.9e-3); 44/123 patients parallel
Early founding trunk (<50%)
Late subclonal branching (>50%)
RTK/RAS & metabolic signaling
Epigenetic / chromatin / splicing
Bubble size & (n): within-patient parallel hits
Pan-cohort independent branch acquisitions
NPM1 (3)
NRAS (7)
FLT3 (8)
RUNX1 (7)
TET2
DNMT3A (5)
ASXL1 (3)
SRSF2 (4)
IDH2
KRAS (3)
KIT (3)
IDH1
PTPN11 (3)
WT1
TP53
SETBP1
JAK2
U2AF1
STAG2
Subclonal late branch acquisition (% of patient trees)